

3. Channels of technology transfer and intellectual property rights in developing countries

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1. INTRODUCTION

This chapter presents an empirical analysis of the influence of the strength of intellectual property rights (IPRs) on technology transfer to developing nations. The core contribution is to analyze the relationship between various measures of technology transfer and a set of indexes that quantify the strength of IPRs based on case laws and statutes, while controlling for other factors. The modes of technology transfer examined include services trade (including the licensing of intangible assets), merchandise trade, and foreign direct investment (FDI). The study also evaluates the relationship of IPR strengthening and the associated technology transfer to the evolution of local innovative capacity of developing countries using data on research and development (R&D) and patenting by residents and non-residents. For the analysis the authors have assembled a large data set covering a broad international panel of countries for the time frame 1990 to 2005. The time frame and country coverage are significantly expanded in comparison with previous studies on IPRs.

A central issue in academic and policy debate is not only the influence of IPRs on technology transfer but on whether IPRs induce *technologically-intensive* transfers. In other words, how substantive are the induced technology transfers, if any? It is possible that IPR reforms primarily induce foreign investment in wholesale trade operations or in low wage manufacturing facilities. This chapter evaluates the technological content of technology transfers in several ways. First, it examines the sectoral impacts – to determine whether inward FDI and trade arise in relatively high-tech sectors (rather than say wholesale trade, finance, tourism, and holding companies). Second, it examines the associated changes in foreign patenting. If firms transfer technologically sensitive assets, they will likely file for patent protection for them. In the event that firms transfer outdated

technologies or non-technologically intensive assets, there is less demand for patent protection.¹ Third, the paper examines whether inward technology transfers influence local innovation. To the extent that these inflows have substantive technological content, they will be useful inputs into local innovation. Based on these examinations, the chapter finds overall that substantive technology transfers are associated with stronger IPRs in developing economies.

The chapter is organized as follows. Section 2 provides a review of the empirical literature on technology transfer and the role of IPRs, and discusses gaps in the existing work. Section 3 sketches the empirical framework of the study (with further technical details provided in Appendixes 3.1 to 3.3). Section 4 presents the data set, including the measures of IPR strength. Section 5 presents the empirical results in two parts. The first focuses on assessment of which types of IPRs are most related to the different types of technology transfer, and whether the relationship of IPRs to technology transfer varies by the level of economic development in the IPR reforming countries. The second part of Section 5 focuses on the high-technology content of the imports and inward FDI, in order to consider whether the flows of this content are related to R&D and patenting in the developing world. Section 6 concludes with a summary of the main findings, implications, and suggestions for further research.

1.1 Motivation

The motivation for this study is to shed further light on the extent to which developing nations gain from IPR reform. Intellectual property systems require resources. In light of that, it may be tempting to argue that developing economies have an incentive to avoid the costs of IPR systems by copying or free riding on the innovations of industrialized economies. However, this argument ignores the consequences of a lack of an IPR system on inward technology transfers. Technology transfers, such as foreign direct investment (FDI), are a source of capital, technology, skills, management techniques, and employment opportunities. Moreover, foreign investment, imports, and licensing can help generate competition in the host country.

The intellectual property regimes of developing countries have been significantly strengthened since 1990, notably through the advent of the World Trade Organization's Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS), which came into force in January 1995. Further impetus to this wave of IPR reform came about as a consequence of economic transition in formerly socialist countries. Reform of IPRs remains an ongoing process globally, including in the developing world.

Intense public debates sometimes arise about the efficacy and desirability of further IPR reform in developing countries. Theoretical analyses sometimes provide conflicting views on the role of IPRs in economic development.² This situation highlights the need for empirical evaluations of the effects of IPR reform to date, in order to anchor future discussions and policy making with findings from actual experience. Previous empirical work ranges from informal, anecdotal evidence to formal case studies and regression-based statistical analyses. The formal evidence, however, is still scant and not definitive, particularly as regards the extent to which IPR reforms in developing countries help substantively attract foreign technologies or spur local innovation.

A key constraint in a number of previous empirical studies is the lack of up-to-date measures of the progress of global IPR reform. Much of the existing empirical analysis is based on data for the pre-TRIPS period; that is, the studies were largely conducted on data sets where the time period ended before 1995. Of course, lags in the generation of data for such indicators as FDI, trade and R&D are to be expected, but up-to-date quantitative measures for IPR strength have lagged even further. This study fills a gap by updating and extending indexes of IPR strength up to 2005, thereby incorporating recent developments in international IPR protection. This enables the whole sample period to be brought up to 2005, and allows more recent data on technology transfer to be used than was the case in previous studies.

Basing policy on empirical relationships derived for standards of protection that prevailed prior to the TRIPS Agreement is somewhat problematic since changes in the policy regime are likely to have affected public expectations and behaviour. The elasticities or responsiveness of innovation and technology transfer, for example, may be quite different under different IPR regimes. Moreover, TRIPS is a relatively recent agreement and studies attempting to evaluate the effects of IPRs based on the first few years of data may have missed substantive effects, especially where technology transfer and technological changes respond to IPR reforms with a lag.³ Some time needs to pass in order to observe fundamental rather than transient impacts. The present study has at least a ten year window of observation since the TRIPS Agreement came into effect.

2. LITERATURE REVIEW

2.1 Developing Country Perspectives on IPRs

Developing country perspectives vary on the importance of IPRs as a component of economic policy. Public debate on IPRs in these countries

is sometimes caught up in emotive issues such as implications for public health and access to medicine⁴ or the need to prioritize among many competing demands for limited government resources. Critics point to significant implementation costs that can be associated with IPR commitments undertaken in the various international agreements.^{5,6} Correa (2005) and others have challenged the legal and economic implications of strengthening of IPR, alleging that the system of international IPR rules is imposing a burden on developing countries. The accusation is that the emerging standards raise the cost of intellectual content in products sought by developing countries, while developing countries may not have the capacity to capitalize on their own potential in a similar manner. Moreover, Correa alleges that the implicit bargain underlying the strengthening of the international IPR regime has not been satisfied. Some developing countries have argued in the World Trade Organization (WTO) and World Intellectual Property Organization (WIPO) that promises of technology transfer as contained in the TRIPS Agreement (see Box 3.1) do not appear to be yielding corresponding benefits for developing countries, whereas the strengthened IPR may raise costs for developing countries seeking to upgrade their technological capabilities.⁷

On the other hand, some developing countries have sought to exploit strengthened IPRs strategically as a development tool. For example, they may see IPR policy as a means to draw in investment and encourage domestic innovation, with the potential to boost development on both the extensive dimension (overall size of the economy) as well as the intensive dimension (i.e. the value-added per employee). Government officials from a range of economies have pointed to strengthened IPRs as a plank in their strategies to enhance FDI inflows and trade.⁸ For example, experts in some poor developing countries have seen the institution of trademark protection as a vehicle for reassuring investors in manufacturing industries that they can combat knock-offs. For wealthier countries, enhancement of IPRs may be seen as a means to draw in high technology that can boost worker productivity and contribute to intensification of growth.

2.2 Technology Transfer

The empirical economic literature is increasingly tending to lend support to the view that international trade and FDI provide means by which technologies can be spread internationally (e.g. see Keller 2004 for a survey). Acharya and Keller (2007) show how the extent of international technology transfer activities can explain cross-country differences in per capita incomes. They find that imports are an especially important channel of technology transfer, although the volume of transfer varies by country,

BOX 3.1 THE TRIPS AGREEMENT AND TECHNOLOGY TRANSFER

The TRIPS Agreement makes explicit the signatories' intention to promote technology transfer. Article 7 includes a corresponding reference as part of the general objectives of the Agreement:

The protection and enforcement of intellectual property rights should contribute to the promotion of technological innovation and to the transfer and dissemination of technology, to the mutual advantage of producers and users of technological knowledge and in a manner conducive to social and economic welfare, and to a balance of rights and obligations.

Moreover, in the provisions governing patent rights the Agreement provides for a mechanism for rights holders to transfer their property for use by others:

Patent owners shall also have the right to assign, or transfer by succession, the patent and to conclude licensing contracts. (Article 28.2)

The Agreement includes other provisions aimed at preventing abuses that would limit technology transfer. For example, Article 8.2 states the principle that:

Appropriate measures, provided that they are consistent with the provisions of this Agreement, may be needed to prevent the abuse of intellectual property rights by right holders or the resort to practices which unreasonably restrain trade or adversely affect the international transfer of technology.

Article 40 addresses issues related to control of anti-competitive practices in contractual licences, specifying the approach to dealing with this issue, beginning with the following point of understanding:

Members agree that some licensing practices or conditions pertaining to intellectual property rights which restrain competition may have adverse effects on trade and may impede the transfer and dissemination of technology.

Article 66.2 provides for incentives to encourage technology transfer from developed countries to LDCs:

Developed country Members shall provide incentives to enterprises and institutions in their territories for the purpose of promoting and encouraging technology transfer to least-developed country Members in order to enable them to create a sound and viable technological base.

At the WTO Ministerial Conference in Doha in November 2001, ministers issued a decision on Implementation-related issues and concerns that elaborated upon Article 66.2 (WTO document WT/MIN(01)/17, para 11.2):

Reaffirming that the provisions of Article 66.2 of the TRIPS Agreement are mandatory, it is agreed that the TRIPS Council shall put in place a mechanism for ensuring the monitoring and full implementation of the obligations in question. To this end, developed-country members shall submit prior to the end of 2002 detailed reports on the functioning in practice of the incentives provided to their enterprises for the transfer of technology in pursuance of their commitments under Article 66.2. These submissions shall be subject to a review in the TRIPS Council and information shall be updated by Members annually.

In its Decision of 19 February 2003 (WTO document IP/C/28), the Council for TRIPS put in place the reporting mechanism which is now operational. Developed countries submit annual reports on their actions in relation to Article 66.2. A review of the annual reports at the October 2006 meeting of the Council for TRIPS, however, revealed continuing divergence between some developed and developing countries in the interpretation of Article 66.2. The minutes from this meeting (WTO document IP/C/M/52) and other Council documents highlight differences with respect to the definition of “technology transfer”, the ability to measure “technology transfer” and the role of both developed and developing countries under the Agreement, among other issues. For example, there is debate as to whether technical assistance related to implementation of the TRIPS Agreement can be considered as contributing to satisfaction of Article 66.2.

being greater in countries with stronger absorptive capacities (as measured by the level of local education attainment and research and development investments). Other studies on imports as a channel of transfer focus on producer goods and intangible assets. For example, Eaton and Kortum (2002) focus on imports of intermediate inputs and show that trade can

increase the productivity of importing countries by increasing their access to foreign inputs and technologies. Coe and Helpman (1995) focus on spillovers of research knowledge transmitted through trade; that is, the extent of spillovers from country X into country Y depends on the share of X's products in the total imports of country Y. Xu and Wang (1999) argue that it is the share of capital goods imports that matters most in conveying knowledge spillovers.

A portion of the literature concentrates on FDI as a means of international technology transfer. Hoekman and Javorcik (2006) provide cross-country as well as country studies of FDI as a conduit for knowledge transfer. Such transfers help recipient countries develop their own capacity to export high-tech goods and learn-by-exporting. Other studies explore the mechanisms by which FDI transfers knowledge to the local economy. For example, the transmission of technology can occur when technology is transferred from the parent firm to the subsidiary (see Markusen, 2002) or when technological goods produced by subsidiaries find uses locally (see Blomström and Kokko, 1998). The transmission of technology or knowledge can also occur through labour mobility between subsidiaries and domestic enterprises (Fosfuri et al., 2001).

Although there is mounting evidence that FDI and imports can be vehicles for technology transfer, the studies referenced above do not explicitly examine the role of IPRs in stimulating technology transfer via trade and FDI. Maskus (2004) provides a more focused survey of IPR policies and regimes, noting in particular that they affect incentives for engaging in international technology transfer, influencing the volume of flows via different mechanisms of transfers as well as the choice of means of transferring technologies.

Furthermore, the literature points to complexity in the influence of IPRs on such processes. A priori, IPRs create possibilities for both expanding and restricting technology transfer. For example, Maskus and Penubarti (1995) identify two opposing theoretical effects of stronger IPRs on trade: *a market expansion effect* and *a market power effect*. These effects could also be present in other forms of technology transfer activities. Consider a firm in country A that exports patentable commodities to country B, and suppose country B strengthens its IPRs. On the one hand, the firm perceives an expansion in its market due to a reduction in imitation by local firms. The demand curve it faces in country B shifts out. On the other hand, stronger IPRs in country B increase the firm's market power, reducing the elasticity of the demand it faces. The market expansion effect is likely to dominate in countries where the degree of market rivalry and competition is high, but the market power effect could in theory prevail in countries where local competitors pose a weak threat of imitation.

2.3 Trade and IPRs

Although a number of studies have considered the influence of IPRs on trade flows, the evidence is far from complete. Maskus and Penubarti (1995) use data for 1984 and examine manufacturing exports from OECD member countries (as of 1984) to developing nations. Empirically they find that the market expansion effect tends to dominate the market power effect so that stronger IPRs on balance expand trade. Fink and Primo Braga (1999), using a sample of 89 countries in 1989, find a positive link between IPRs and trade flows for total non-fuel trade but a weak link between IPRs and high technology trade (such as trade in chemicals, electrical and office machinery, telecommunications apparatus, and so forth). One reason may be that the market power effect of IPRs dominates in high-technology sectors. That is, firms in technology-intense industries may already possess some market dominance from their technological superiority. Stronger patent protection would then largely raise their rents. Another possibility, and one that affects many studies of technology transfer that focus on one mode of technology transfer, is that stronger IPRs lead firms to switch among different modes of transfer, say from exporting to FDI.⁹

However, later studies do find that stronger IPRs, in certain situations, can stimulate trade in high-tech products. Using state-level US exports to 92 countries for the period 1992, Smith (1999) finds that the impact of stronger patent protection on trade depends on the importing country's capacity to imitate (using R&D/GDP ratios as an indicator of the capacity to imitate). In countries with a weak capacity to imitate, the weak capacity acts as a de facto form of protection. Firms would not actually need IPRs to appropriate the returns to their innovation. Thus in such circumstances stronger patent rights provide greater market power, which could outweigh any market expansion effects. The empirical results in Smith (1999) confirm that where the capacity for imitation is high, a strengthening of IPRs increases the flow of trade. This holds in high-technology sectors as well, such as chemicals and scientific instruments.

IPRs alone may not stimulate knowledge transfers. Bascavusoglu and Zuniga (2002) examine the exports of French technology services to 19 country destinations and 29 sectors for the year 1999. They find that foreign IPRs can enhance such exports if the destination markets have commercial potential in terms of large markets and strong technological capacities. Stronger IPRs in low-income countries have an insignificant effect and may even deter technology flows in low-technology sectors.

In all of the studies cited above, cross-sectional data were used. Co (2004) uses panel data (i.e. cross-sectional and time-series data) that allow for dynamic changes in IP institutions and market structure. The data set

covers US exports to 71 countries during the period 1970–92. The study finds that stronger IPRs stimulate US exports of goods that are R&D-intensive (such as drugs and electronics) conditional on the importing country having at least a moderate capacity to imitate. Stronger IPRs are found to reduce US exports of non R&D-intensive goods to countries with a low capacity for imitation, holding other factors constant.

2.4 FDI and IPRs

Empirical evidence on the effects of IPRs on FDI is rather mixed. Primo Braga and Fink (1998) examine the stocks of outward US direct foreign investment in 42 countries in 1992 and find that they weakly correlate with the strength of patent protection in those countries, holding other factors constant. In contrast, Lee and Mansfield (1996) examine a panel of 14 developing countries around the same period studied in Primo Braga and Fink (1998) and find that the strength of IPRs (as perceived by managers in the Mansfield (1994) survey) is a significant determinant of the volume of US outward FDI flows. Lee and Mansfield (1996) also find that weaker IPRs can affect the composition of FDI, causing firms to invest in non-manufacturing and non-R&D activities, like sales and distribution outlets.

More recently, Nunnenkamp and Spatz (2004) using aggregate national data find that intellectual property rights are a significant determinant of US outward FDI stocks, particularly in developing countries. Branstetter et al. (2005) concur, using firm level data in 16 countries during the period 1982–1999. They find that host country IPR reforms encourage US multinational firms (especially those parent firms with large patent portfolios) to expand their sales, employment, investment, and production abroad.

In contrast, Mayer and Pfister (2001) using French multinational firm data for the periods 1981–83 and 1988–92 find that stronger patent rights have a negative influence on the location decisions of multinational firms. When they split their sample into developed and developing host countries, they find that the strength of a developing country's patent laws has a statistically insignificant influence on the probability that a French multinational firm will locate in that developing country. The strength of a developed country's patent laws has a quadratic (inverse-U) effect on the firm's probability of locating in the developed country; that is, increasing the probability and then decreasing it after some critical level of patent law strength is reached.¹⁰ It is important to note that Mayer and Pfister (2001) study location decisions, not FDI flows or stocks. For firms already located in a country, the intensity of technology transfer in response to changes in patent laws is not captured in the location data alone.

Using firm-level data for Eastern Europe and the former Soviet Union

states, Javorcik (2004) finds that stronger patent rights have a positive and statistically significant effect on the probability of foreign investment in high-technology sectors and an insignificant effect in other sectors. Moreover, foreign investors are more likely to invest in sales and distribution outlets rather than in manufacturing or R&D facilities when patent protection is weaker; this propensity is found in all sectors, not just in high technology. These findings conflict with those of Mayer and Pfister (2001) but are consistent with those of Lee and Mansfield (1996).

Thus while the evidence on the effects of IPRs on FDI is mixed, studies using sectoral or industrial data seem to find that technology-sensitive FDI does respond to developing country IPRs. Moreover IPRs seem to affect the functional composition of FDI, namely between sales and distribution outlets and production and R&D facilities.

2.5 Gaps in the Literature

The present study differs from the previous literature in several respects. First, previous studies have largely covered the time period before the TRIPS agreement went into effect. The levels of IPRs did not change significantly before then. Thus previous studies depended mostly on cross-country variations in IPRs to explain differences in technology inflows across countries. Moreover, the structure of economies, globally and nationally, may have shifted in the post-TRIPS era (e.g. regime shifts), so that it would be useful to estimate relationships using more recent data. This study examines the period 1990–2005 and utilizes up-to-date measures of IPRs.

Second, previous studies have not separated merchandise trade and services trade. Trade in services has expanded along with increases in world FDI. In effect FDI is a conduit for “exporting” services. Another reason to distinguish between services and merchandise trade is that technology transfer can consist of *tangible goods* (such as technological machinery) made available through merchandise trade, and *intangible goods* (such as technological know-how) made available through services trade. Indeed, Miroudot (2005) discusses how increasingly innovative the services sector has become and how trade in services is an important vehicle for knowledge diffusion.

A third aspect that distinguishes this study from previous studies is that more attention is paid to ascertaining the technology content of imports and inward FDI into developing countries. A key issue in the international technology debate is whether trade or FDI actually introduces new technologies to developing countries or just older vintages of technology.¹¹ In the case of FDI, plants were often established to serve as outlets

for sales and distribution or for low-value added manufacturing, rather than centers for research and high-value added manufacturing. This may be understandable if developing economies had weak IPR systems and posed risks of imitation. Indeed, Contractor (1981) presents evidence that US firms tended to transfer older technologies to unaffiliated parties in developing countries compared to those transferred to agents in industrialized economies.¹² Mansfield's (1994) survey provides concurring evidence that technology-intensive firms are reluctant to transfer new technologies to countries with weak IP laws. More recently, Nunnenkamp and Spatz (2004) find that weaker IPRs are associated with lower quality FDI as judged by the small increases in local R&D, employment, and value added that accompany the FDI. Further evidence on the technology transfer dimension of these flows is provided in Branstetter et al. (2006). Their analysis of US multilateral firms' responses to IPR reform in 16 economies during 1982-99 "reveals that royalty payments for technology transferred to affiliates increase at the time of reforms, as do affiliate R&D expenditures and total levels of foreign patent applications." The findings point to prospects of increased technology transfer from US firms to affiliates in countries that engaged in IPR reform.

In this study, the response of technology-intensive imports and inward FDI to the strength of IPRs is examined. The study also examines the response of local innovation and technology diffusion to IPRs and inward technology transfer.

3. CONCEPTUAL FRAMEWORK

The core contribution of this study is to present evidence, based on regression analysis, of the impact of IPRs on technology transfer to developing countries.¹³ This section provides a brief sketch of the empirical strategy. The technical details are relegated to Appendixes 3.1 to 3.3.

The primary objective is to analyze the relationship between technology transfer and intellectual property rights, controlling for other factors:

$$\text{Technology Transfer} = f(\text{IPR, Control Variables}) + \text{error term} \quad (3.1)$$

where f denotes "function of" and IPR intellectual property rights. The error term is decomposed into a country-specific, time-invariant effect that captures country-specific omitted variables and a random error term that captures the idiosyncratic heterogeneity (or volatility uncorrelated to the other variables). Unless indicated otherwise, we take the natural logarithms of the dependent and independent variables.

The measures of technology transfer considered are inward FDI, merchandise imports, and service imports. The types of intellectual property rights considered are patent protection, copyrights, and trademark protection. The study also looks at the enforcement of IPRs in general by using a survey variable which measures the perception of IPR strength and experiences of business executives in different countries.

To ensure that the empirical analysis picks up the relationship between IPRs and technology transfer, we control for variables that may also influence technology transfer, as well as variables whose influence on technology transfer the IPRs may be picking up instead. For example, strong institutions or property rights (to physical property) are correlated with levels of IPRs. Thus, unless those institutional features are controlled for, the IPR variable may simply be a proxy for institutional quality or for property rights in general. That is, if a positive relationship between IPRs and technology transfer is picked up, the relationship may be due to the role of institutions or property rights per se rather than IPRs. Hence it is important to control for these other institutional features.

The control variables for the study include indicators of property rights in general, legal effectiveness, governance, the business environment, the nature of the trade regime, and per capita GDP. When assessing the response of local innovation to IPRs and technology transfer, the study also controls for the quality of research institutions and the degree of university and industry research collaboration.

A key question in the global IPR debate is the extent to which stronger IPRs enable developing countries to acquire new technologies. Thus the next exercise is to assess the technology content of imports and FDI into developing and least developed economies. Two approaches are examined. First, equation (3.1) above is re-estimated using sectoral or industrial level data. In particular, the inward FDI and imports of high-tech sectors or industries, such as pharmaceuticals, chemicals, computers, and others, are related to the strength of IPRs in the developing world. The empirical estimates would indicate the extent to which IPR reform attracts technology intensive imports and capital, holding other factors constant.

A second approach is to relate measures of innovation in the developing world, such as R&D and patenting, to measures of IPR and technology transfer:

$$\text{Innovation} = f(\text{IPR, Technology Transfer, Control Variables}) + \text{error term} \quad (3.2)$$

The intuition is as follows: if inward FDI or imports are associated with new technological products or processes, foreign producers would have an

interest in applying for local patents to protect against copying.¹⁴ Thus a rise in technology-intensive imports or inward FDI would be associated with a rise in non-resident patenting, holding other factors constant. In contrast, if foreign producers' exports to, or FDI in, the developing world were low in technology content or consisted of outdated technology, firms would less likely file for patent protection. Thus a statistically significant rise in non-resident patenting could reflect the degree to which imports or inward FDI introduce new technologies to developing markets.

Local or domestic R&D and patenting could also be affected by the technological content of imports and inward FDI. Imports or foreign direct investments that are high in technology content have the potential to generate knowledge spillovers, benefiting local researchers and innovators. Moreover, technology-intensive imports or inward FDI provide useful equipment (such as laboratory capital) and materials (such as chemical ingredients, biological samples) to aid research and innovation. In contrast, if developing country imports and inward FDI are low in technological content, such inflows would insignificantly contribute to local R&D and patenting. Hence a statistically significant rise in local R&D and resident patenting could reflect the degree to which imports or inward FDI are high in technology content. Thus to the extent that IPRs stimulate technology transfer, IPRs can indirectly affect local innovation by increasing access to new knowledge and directly by affecting incentives to innovate.

These potential effects of IPRs can be summarized in the following two hypotheses:

Hypothesis 1: IPRs stimulate technology transfer, particularly the transfer of technology-intensive goods, services, and capital.

Hypothesis 2: IPRs can directly stimulate local innovation as well as indirectly by stimulating the transfer of technologies that foster local innovation.

The goal of the empirical work is to evaluate these two hypotheses for developing countries and least developed countries (LDCs). Of course, good theoretical reasons also exist as to why IPRs could in some instances hinder technology transfer (e.g. where the market power effects of IPRs dominate the market expansion effects) or innovation (e.g. by increasing the costs of research inputs, by creating patent thickets which require costly licensing negotiations to resolve, or by blocking important technological components needed by other innovators). Thus, the hypotheses above concern the *net* empirical effects of IPRs.

There are also limitations with the use of patent data to measure innovation and technology diffusion. Not all new innovations are patented.

Some firms may prefer to keep new inventions a trade secret or may have developed innovations that are not patentable in some jurisdictions such as software, biotechnological innovations, or business methods. Other firms may not bother to apply for patent protection in jurisdictions where the capacity for imitation is weak. Thus drawing conclusions about the technology content of imports and FDI based on variations in patent filings must be subject to these qualifications.

Note that our independent variables may consist of time-invariant variables. Also, due to the fact that the countries in the sample are quite heterogeneous – for example, of different economic sizes – the variance of the error term is unlikely to be constant across countries but to vary with size. In other words, the error term is likely to be heteroskedastic (e.g. have unequal variances across units). Failure to take this into account means that the estimated model would give greater weight to “larger” countries (i.e. countries with larger error variances) than to smaller. Both equations (3.1) and (3.2) are estimated by the method of generalized least squares (GLS) to adjust for such nonconstant error variances. More specifically, they are estimated by *Feasible GLS*, as the form of heteroskedasticity must be estimated prior to applying GLS.

4. DATA

This section describes the dataset, data sources, and some sample statistics.

4.1 Dependent Variables

A large sample of countries has been assembled for this study, covering the time period 1990–2005. The sample of countries is diverse, representing different income groups and institutional environments (see Appendix 3.1 for a list of countries and Appendix 3.2 for a list of data sources).

As measures of inward technology transfer, the study employs the stock of inward FDI, merchandise imports, and service imports.¹⁵ Data on inward FDI and services trade (including services by sector) are from UNCTAD, while data on merchandise trade are from the WTO. This study also looks at the sectoral or industrial composition of inward FDI and imports of merchandise goods and services. Data on merchandise trade by sector are from the WTO and the UN *Comtrade* database, while data on services trade by sector are from UNCTAD. Sectoral data on global FDI are not widely available. Hence, as a substitute, this study examines US outward FDI-related data by sector. Data on US direct investment abroad are from the Bureau of Economic Analysis (BEA). The dependent

variables for FDI and trade are given in constant (2000) US dollars and included as levels.

In addition to measures of technology transfer, the study examines measures of innovation such as R&D and patenting. R&D can be considered an “input” into innovation whereas patents can be considered “outputs” of innovation. The R&D data employed here are from the OECD and UNESCO, and the patent data are from WIPO.

4.2 Explanatory Variables

As explanatory variables, the key independent variables of interest are the strength of IPRs. Four measures of IPRs are considered in this study: an index of patent rights, index of copyrights, index of trademark rights, and a survey rating of IPRs. The first three indexes are based on membership in corresponding international treaties (which influence IPRs, see Box 3.2), statutory laws and legislation, as well as case law. The fourth measure is based on the perceptions and experiences of business executives in different countries. A description of each measure is provided below, and Appendix 3.3 provides a summary of the legal features in the indexes of patent rights, copyrights, and trademark rights.

4.2.1 Patent Rights Index

The measure of patent rights is based on Ginarte and Park (1997) and Park (2008). This present study updates the index to 2005 and extends it in order to incorporate the patentability of new technological fields like software and biotechnology. The index of patent rights ranges from 0 (weakest) to 5 (strongest). The value of the index is obtained by aggregating the following five components: extent of coverage, membership in international treaties, duration of protection, absence of restrictions on rights, and statutory enforcement provisions. Coverage refers to the subject material (type of invention) that can be protected; duration refers to the length of protection; restrictions refer to the less than exclusive use of those rights; membership in international treaties indicates the adoption into national law of certain substantive and procedural laws of those international agreements. Membership in an international treaty may also signal the willingness of particular nations to adhere to shared international principles such as non-discrimination. The enforcement component consists of mechanisms that aid in enforcing one's patent rights (such as preliminary injunctions against infringers). Each of these components is scored on a scale from 0 to 1 (reflecting the fraction of legal features that are available). The overall value of the patent rights index is the unweighted sum of the component scores.

BOX 3.2 TRIPS AND RATIFICATION OF WIPO-ADMINISTERED TREATIES

WIPO administers a series of international IPR treaties developed over many years. Since 1990, the geographic coverage of these treaties has been notably extended via increased numbers of ratifications. This occurred, in particular, in relation to the launching of economic reforms in the former socialist countries and in association with the coming into effect of the TRIPS Agreement (which references the Bern, Paris and Rome Conventions). The increase in ratifications is especially striking for the Berne and Paris Conventions (see Figure 3.1).

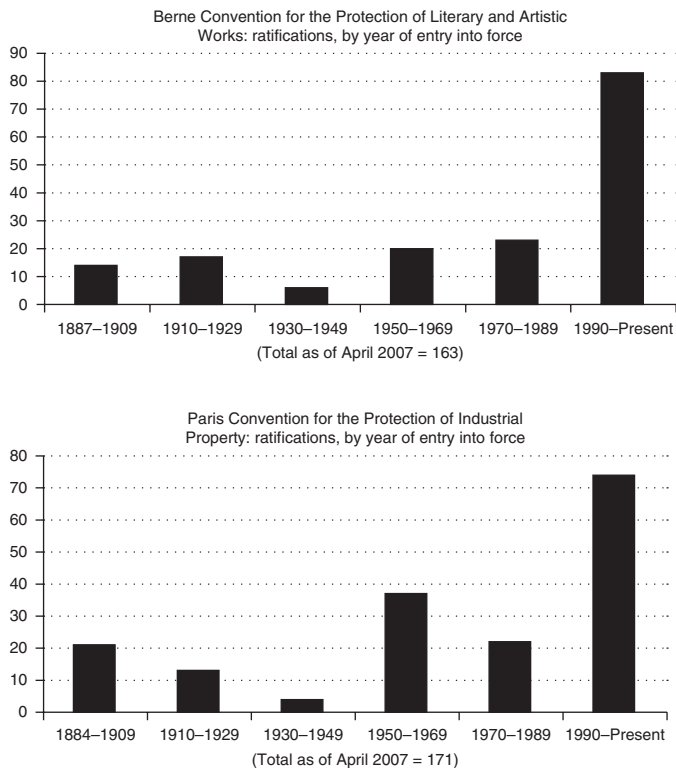


Figure 3.1 Selected WIPO conventions referenced in the WTO TRIPS Agreement

The Berne Convention for the Protection of Literary and Artistic Works (ref. copyrights and neighbouring rights) attracted 59 new ratifications (as of October 2007, the total number was 163); The Paris Convention for the Protection of Industrial Property (ref. patents, trademarks, industrial designs, utility models, geographical indications) attracted 60 new ratifications (as of October 2007, the total number was 172); The Rome Convention for the Protection of Performers, Producers of Phonograms and Broadcasting Organizations experienced 30 new ratifications (as of 20 October 2007, the total number was 86).

The increase in ratifications extended specific protections for the categories of intellectual property covered by the conventions; these rolled out as countries moved to comply with the accords. For example, all three conventions include provisions on “national treatment”, meaning that within the terms specified in the conventions, the ratifying country must offer non-nationals the same protection as it grants to its own nationals. Other provisions specify the handling of various dimensions of IPRs, such as the minimum duration of protection (Berne and Rome Conventions), nature of the process for extension of rights (e.g. automatic for copyrights in the case of the Berne Convention and by extension the Rome Convention, case by case on a national basis for patents in the case of the Paris Convention), and rights during the application process (e.g. right of priority specified under the Paris Convention), among other provisions.

4.2.2 Copyrights Index

This index is based on Park (2005) and Reynolds (2003), and the present study updates the index to 2005. The copyright index consists of four components: coverage, usage, enforcement, and membership in international treaties. Coverage again refers to the subject matter that is protected and is intertwined with copyright duration (since the length of protection varies with subject matter). The usage component addresses the degree to which copyright holders have control over their copyrights (vis-à-vis the use of their works by others). The enforcement component also includes provisions that aid in enforcing a copyright holder’s rights (such as the availability of criminal penalties for infringement). The treaties cover

various global conventions and agreements (listed in Appendix 3.3). Each component is scored on a scale from 0 to 1, again reflecting the fraction of legal features that are available. The overall score for the copyright index is the unweighted average of the four components. Hence, the copyright index ranges from 0 (lowest) to 1 (highest).

4.2.3 Trademark Rights Index

This index is also based on Park (2005) and Reynolds (2003), and for this study the index is updated to 2005. The trademark index consists of three components: coverage, procedures (which incorporates enforcement features and possible restrictions on the rights holder), and international treaties. The coverage component refers to types of names and symbols that can be trademarked. The procedures component addresses the manner in which trademark rights are procured and enforced; hence the procedures component incorporates enforcement features. The international treaties component incorporates various global conventions and agreements on statutory and procedural laws. Each component is scored on a scale of 0 to 1, indicating again the fraction of legal features that are available. The overall score for the trademark index is an unweighted average of the three components.

4.2.4 IPR survey

This index is from the World Economic Forum surveys of business executives. Business executives in the country surveyed are asked to rate the country's strength of intellectual property laws on a scale of 1 (weakest) to 7 (strongest). Their ratings are based on their experiences and perceptions, as well as on any information or observation they may have of the laws, procedures, and institutions. This makes the survey ratings a useful complement to the statutory indexes above. Nonetheless, there are a few limitations with this index. First, it is highly aggregative in terms of not distinguishing among copyrights, trademark rights, or patent rights. It is difficult to know which aspects of IPR laws or practice drive the ratings to be high or low. Second, the ratings of different countries are not fully comparable because the same group of survey participants is not evaluating the full sample of countries. Rather, participants in country A rate country A's IPRs laws, while participants in country B rate country B's IPRs laws. Participants in A may have stricter standards than those in B. Hence the ratings are less comparable than if the same participants were to rank country A versus country B. Third, the time series dimension is limited. Data covering a large number of countries are available for 2004–07, but a smaller sample of countries is available for 2000. Due to the limited time dimension of this variable, this study takes the average values for 2004–05 and treats the data as a time invariant variable).

4.2.5 Control variables

A number of factors other than IPRs could affect technology transfer. This study controls for physical property rights in general, the effectiveness of the legal regime, quality of governance, the cost of doing business, the freedom to engage in trade, and per capita GDP. A number of factors other than IPRs could also affect local R&D and patenting. The study controls for the above mentioned institutional variables as well as the quality of research institutions and the degree of university and industry research collaboration. Data on institutional controls are from the World Economic Forum and the Economic Freedom Network (EFN). These organizations conduct surveys internationally and pool the survey opinions to form indexes. In some cases, quantitative data are used as inputs into the creation of the indexes; for example, the EFN uses tariff data and levels of capital controls to help determine an index of free trade. The rest of the data – on governance, GDP per capita, and business environment – are from the World Bank.

4.3 Sample Statistics

Particularly since the entry into force of the TRIPS Agreement in 1995, developing and least developed countries have undertaken significant reforms in their intellectual property systems. The indexes of IPRs show that between 1995 and 2005, developing and least developed countries, as a group, experienced a greater percentage increase in IPR strength than did the developed world. Figure 3.1 illustrates this with respect to the Patent Rights Index. As Table 3.1 shows, the strength of IPR laws varies positively with the level of economic development.¹⁶ Group 1 (developed countries) has the strongest patent, copyright, and trademark systems, followed by Group 2 (developing countries) and then by Group 3 (LDCs). Business executives surveyed also rate the intellectual property regimes in Group 1 as the most stringent, followed by those in Group 2 and then Group 3.

While levels of IPR strength are highest among developed countries, the most significant IPR reforms since 1995 have occurred among developing and least developed countries. As Table 3.1 also shows, the percentage changes in the IPR indexes are larger for Groups 2 and 3 than for Group 1, except in the case of the copyright laws of Group 3. Protection levels for artistic creations have hardly changed since 1995. This may be due to the weak presence of copyright industries in LDCs compared to enterprises that exploit trademark and technological assets. Table 3.1 also shows that the different IPR indexes are all positively correlated with one another; the IPR survey variable is most correlated with the strength of patent rights.

Table 3.2 presents some trends in the measures of international

Table 3.1 Intellectual property rights (IPR) sample statistics

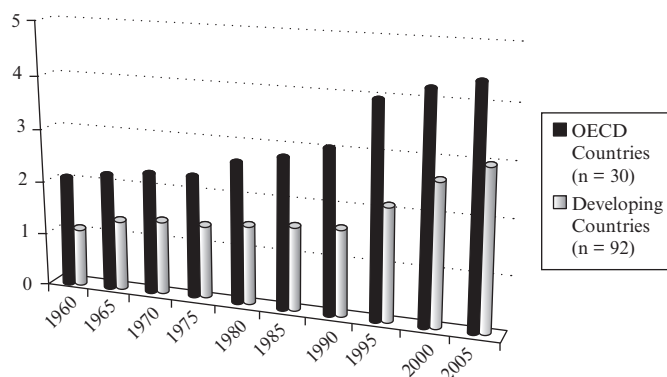
Part A:				Correlations	
	Index of Patent Rights Group Mean in 2005 (% since 1995)	Index of Copyrights Group Mean in 2005 (% since 1995)	Index of Trademark Rights Group Mean in 2005 (% since 1995)	IPR Survey Group Mean in 2005 (% since 1995)	
Group 1	4.4 (10%)	0.8 (14.3%)	0.7 (16.7%)	5.5 (n/a)	
Group 2	3.3 (37.5%)	0.6 (20%)	0.6 (20%)	3.5 (n/a)	
Group 3	2.4 (26.3%)	0.42 (0%)	0.36 (33.3%)	2.7 (n/a)	
Part B:				Correlations	
	Patent Rights	Copyrights	Trademarks	IPR Survey	
Patent Rights	1.0				
Copyrights	0.53	1.0			
Trademark Rights	0.57	0.58	1.0		
IPR Survey	0.62	0.36	0.37	1.0	

Notes:

Group 1 consists of developed countries, Group 2 of developing countries, and Group 3 of least developed countries. See Appendix 3.1 for a list of countries in each group.

'IPR Survey' refers to the rating of the strength of national intellectual property rights by business executives in the World Economic Forum's *Global Competitiveness Report* survey.

Each entry shows the within group mean in year 2005 and the percentage growth in the group mean from 1995 to 2005 in parenthesis. The index of patent rights varies from 0 (weak) to 5 (strong); index of copyrights from 0 (weak) to 1 (strong); index of trademark rights from 0 (weak) to 1 (strong); IPR survey ratings from 1 (weak) to 7 (strong). n/a denotes not available. See Appendix 3.3 for more details on the construction of the IPR indices.



Note: The vertical bar indicates the advent of the TRIPS Agreement.

Figure 3.2 Evolution of the Patent Rights Index, 1960–2005, by country group

technology transfer. Group 1 is the largest recipient of merchandise and service imports, foreign direct investment, and patentable innovations, and has the highest rate of research and development (R&D) as a percentage of GDP. However, Groups 2 and 3 have experienced larger percentage changes in inflows of goods, services, and capital. They have also experienced larger growth rates in the rate of R&D to GDP. The question then is the extent to which these developments can be attributed to the significant IPR changes in Groups 2 and 3 discussed above.

Table 3.3 shows the industrial composition of these inflows of FDI and trade in developing and least developed countries, focusing on those industries that tend to be technologically-intensive, for example chemicals, pharmaceuticals, computer and information goods, optics, and others. As noted earlier, data on a sectoral breakdown of world foreign direct investment are not widely available. Therefore this study turns to US FDI data for which detailed breakdown by sector is available. US FDI behavior may not be representative of the world as a whole, but the US does account for the largest share of world FDI. Using UNCTAD data sources (see Appendix 3.2), we can calculate that the US accounts for 20% of the world's outward stock of FDI (during 2003–05).

Table 3.3 compares shares between 1995 and 2005 (except for US FDI, where 2005 is compared to 2000).¹⁷ The shares refer to a commodity group's or sector's share in the given activity; for example, the share of pharmaceutical imports in total merchandise imports or the share of communication service imports in total service imports. The key highlights are

Table 3.2 Trends in technology transfer and innovation

Mode of Transfer	Country Group	Mean Value in 2005	Growth since 1995 (%)
Inward Foreign Direct	1	\$244.9	169.8
Investment (FDI)	2	\$35.8	234.3
Stock	3	\$2.05	262.8
Merchandise Imports	1	\$236.2	56.2
	2	\$47.3	91.4
	3	\$2.2	103.8
Service Imports	1	\$58.4	53.4
	2	\$9.2	70.4
	3	\$0.92	91.1
Research and	1	2.3	13.5
Development	2	0.73	22.1
(as a % of GDP)	3	0.32	72.3
Resident Patent	1	57,577	68.2
Applications	2	18,163	687
	3	n/a	n/a
Non-resident Patent	1	60,862	99
Applications	2	24,939	1419
	3	n/a	n/a

Notes: Group 1 consists of developed countries, Group 2 of developing countries, and Group 3 of least developed countries (see Appendix 3.1). FDI and trade figures are in billions of real 2000 US dollars. n/a indicates not available (due to incomplete data).

as follows: there appears to be a shift in the composition of imports and inward investments away from chemicals to other sectors in the developing world. For example, a greater share of US FDI assets is in the service and computer industries in developing countries. A greater share of merchandise imports in Groups 2 and 3 in 2005 is in pharmaceuticals, electrical and electronic goods, office and telecom equipment, and optics and precision equipment, and a smaller share in chemicals and aerospace. Between 1995 and 2005, communication services imports account for a smaller share while computer services account for a larger share, as do royalties. Royalties pertain to the importation of technological services, and measure the license fees and royalties paid for the use of foreign intangible assets (like intellectual property and know-how). While technologically-intensive goods, services, and capital are relatively small shares of foreign inflows into the developing world, they have expanded considerably in developing markets since 1995.

Table 3.4 examines some of the control variables used in this study. The

Table 3.3 Composition of trade and foreign investment in developing and least developed countries

A. Assets of Foreign Affiliates (FA) of US Multinational Firms in Groups 2 and 3		
Industry	Share 2004 (%)	Share 2000 (%)
Chemicals	2.77	5.13
Machinery	1.53	2.30
Electrical equipment; appliances and components	1.51	0.54
Service (professional, scientific and technical)	2.31	1.47
Computers and electronic products	12.80	12.50
Information	2.38	3.02
B. Merchandise Imports of Groups 2 and 3		
Commodity	Share 2005 (%)	Share 1995 (%)
Pharmaceuticals	1.87	1.76
Office and Telecom Equipment	21.30	16.70
Chemicals	3.71	5.08
Optics and Precision Equipment	4.02	3.05
Electronic and Electrical Equipment	22.50	20.30
Aerospace	7.30	7.40
C. Service Imports of Groups 2 and 3		
Sector	Share 2005 (%)	Share 1995 (%)
Communications	1.56	1.94
Computers	1.86	0.79
Royalties (Intangible Property)	6.35	3.80

Notes: Groups 2 and 3 consist of developing and least developed countries (see Appendix 3.1 for a list of countries in each group). Data on US direct investment abroad for the benchmark year 2004 are still preliminary. The final revision is expected to be released in 2008.

table shows, for 2005, the economic and institutional characteristics of Groups 2 and 3 relative to Group 1. The average per capita GDP of developing countries is 15% of that of developed countries, while the average per capita GDP of LDCs is 1% of that of developed countries. Uniformly, Group 2 ranks lower than Group 1 in terms of the degree to which agents are free to engage in international trade, the business climate, the security of property rights, the integrity of the legal system, governance, and innovative capacity. Group 3 in turn ranks lower than Group 2 in terms of these

Table 3.4 Control variables sample statistics

	Group 1	Group 2	Group 3	Correlation with Index of Patent Rights
Gross Domestic Product per capita (in real 2000 U.S. dollars)	1	0.15	0.01	0.67
Index of Freedom to Trade Internationally	1	0.88	0.71	0.55
Doing Business Rank	1	0.27	0.17	0.34
Index of Legal Effectiveness	1	0.59	0.41	0.57
Index of Physical Property Rights (e.g. Land, Tangibles)	1	0.72	0.61	0.58
Governance	1	0.58	0.42	0.67
Innovative Capacity	1	0.65	0.51	0.68
Quality of Research Institutions	1	0.76	0.62	0.61
Extent of University- Industry Research Collaboration	1	0.71	0.55	0.65

Notes: Group 1 consists of developed countries, Group 2 of developing countries, and Group 3 of least developed countries (see Appendix 3.1 for a list of countries in each group). Group means of the above variables are used in the comparisons. Since the 'Doing Business Rank' is in reverse scale to the other variables (that is, higher ranked places have lower costs of doing business), the inverse of the rank was used for purposes of this comparison. See Appendix 3.2 for data sources.

characteristics. As the sample correlations in Table 3.4 show, strong patent rights tend to exist in countries with strong governance, property rights, business climate, open markets, and effective legal systems.

5. EMPIRICAL RESULTS

In this section, the empirical results using regression analysis are discussed. The main results are in Tables 3.5–3.16. Tables 3.5 to 3.10 provide an overview. Table 3.5 shows how the *stock of inward FDI* relates to the different measures of intellectual property strength, holding other factors constant. Table 3.6 repeats the analysis for different country groups: devel-

Table 3.5 *Inward foreign direct investment (FDI) and intellectual property rights*

Dependent Variable:	Inward FDI Stock (in real 2000 US dollars)			
	(1)	(2)	(3)	(4)
Constant	-9.23*** (1.33)	-7.20*** (1.26)	-10.3*** (2.04)	-12.2*** (2.23)
Patent Rights Index	2.01*** (0.19)			2.82*** (0.33)
Copyrights Index		0.79*** (0.23)		-0.27 (0.31)
Trademark Rights Index			0.48** (0.21)	-0.43 (0.23)
Real GDP per capita	0.96*** (0.06)	1.02*** (0.049)	1.22*** (0.08)	0.94*** (0.11)
Freedom to Trade Internationally	2.52*** (0.47)	2.94*** (0.40)	5.13*** (0.55)	4.79*** (0.58)
Doing Business Rank	-0.027 (0.053)	-0.018 (0.06)	-0.049 (0.126)	0.019 (0.11)
IPR Survey	1.61*** (0.49)	2.38*** (0.42)	4.66*** (0.60)	5.36*** (0.63)
Legal Effectiveness	0.87*** (0.32)	0.029 (0.29)	-0.38 (0.39)	-0.28 (0.42)
Physical Property Rights	0.37 (0.58)	0.28 (0.58)	-3.19*** (0.67)	-3.48*** (0.71)
Governance	-1.69*** (0.19)	-1.47*** (0.17)	-1.52*** (0.25)	-1.64*** (0.25)
Number of Observations	261	239	182	170
Pseudo-R ²	63.1%	57.8%	56.9%	62.3%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. Standard errors are in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels respectively.

oped, developing, and least developed. Tables 3.7 and 3.8 do the same for *merchandise imports*, and Tables 3.9 and 3.10 for *service imports*. Having established a link between imports, FDI and IPRs, the next tables indicate the extent to which the inflows of capital, goods and services are associated with effective technology transfer. This study offers two ways in which to assess the technological content of these inflows. The first is a direct approach, namely to assess the sectoral composition of these inflows. Do, for example, IPRs stimulate the inflows of high-tech goods, services and capital? The results of this approach are given in Tables 3.11–3.13. The

Table 3.6 Inward foreign direct investment (FDI) and intellectual property rights by level of economic development

Dependent Variable:	Inward FDI Stock (in real 2000 US dollars)		
	Developed Countries (Group 1)	Developing Countries (Group 2)	Least Developed Countries (Group 3)
Constant	-1.67 (4.25)	-8.73*** (2.37)	-32.7*** (7.14)
Patent Rights Index	11.2*** (1.12)	1.65*** (0.19)	1.66** (0.76)
Real GDP per capita	-0.27 (0.32)	0.99*** (0.099)	3.34*** (0.47)
Freedom to Trade Internationally	2.14 (1.42)	1.78** (0.75)	1.88** (0.78)
Doing Business Rank	-0.19* (0.098)	0.11 (0.13)	3.17*** (1.13)
IPR Survey	6.88*** (1.91)	1.63*** (0.61)	-6.11** (2.64)
Legal Effectiveness	-2.18*** (0.85)	0.70* (0.37)	0.84 (0.60)
Physical Property Rights	-6.19** (2.66)	0.94 (0.72)	4.43* (2.32)
Governance	-0.29 (0.60)	-1.51*** (0.31)	0.10 (0.85)
Number of Observations	67	163	31
Pseudo-R ²	69.6%	35.1%	57.3%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. Standard errors are in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels respectively. Group 1 consists of developed countries, Group 2 of developing countries, and Group 3 of least developed countries; see Appendix 3.1 for the detailed listing of countries.

second is indirect, namely to assess the response of local innovation and technology diffusion to technology transfer. Are increased inward FDI and imports associated with a higher R&D intensity and patenting? The results of this second approach are provided in Tables 3.14–3.16.

5.1 Technology Transfer and IPRs

In Table 3.5, the inward stock of foreign direct investment is investigated. The different IPR measures are examined separately in columns 1–3, and

Table 3.7 Merchandise trade and intellectual property rights

Dependent Variable:	Merchandise Imports (in real 2000 US dollars)			
	(1)	(2)	(3)	(4)
Constant	9.44*** (0.98)	-11.0*** (0.93)	10.5*** (1.70)	8.03*** (1.81)
Patent Rights Index	1.42*** (0.15)			2.15*** (0.26)
Copyrights Index		0.001 (0.12)		-0.96*** (0.26)
Trademark Rights Index			0.37* (0.19)	-0.22 (0.20)
Real GDP per capita	0.83*** (0.042)	0.91*** (0.039)	0.77*** (0.088)	0.61*** (0.11)
Freedom to Trade Internationally	1.05*** (0.31)	1.19*** (0.32)	1.96*** (0.49)	2.21*** (0.42)
Doing Business Rank	-0.065 (0.048)	-0.073** (0.033)	-0.075 (0.072)	0.028 (0.064)
IPR Survey	1.88*** (0.35)	1.92*** (0.39)	3.92*** (0.64)	5.34*** (0.67)
Legal Effectiveness	0.25 (0.19)	0.15 (0.21)	0.63* (0.37)	0.66** (0.33)
Physical Property Rights	0.99** (0.41)	0.49 (0.35)	-1.41** (0.58)	-2.93*** (0.67)
Governance	-1.41*** (0.12)	-1.14*** (0.13)	-1.45*** (0.23)	-1.52*** (0.21)
Number of Observations	264	241	184	172
Pseudo-R ²	64.0%	57.1%	55.6%	60.8%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. Standard errors are in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels respectively.

then jointly in column 4. The results show that individually patent rights, copyrights, and trademark rights are statistically significant determinants of inward FDI. However, when all three measures are entered together, only the patent rights variable remains strongly significant.¹⁸ This suggests that when the index of copyrights or trademark rights is examined alone (along with the control variables), it tends to pick up the effects of patent rights. If so, inward FDI that helps establish plants or subsidiaries is largely sensitive to the protection of inventive output rather than business names, symbols, or artistic creations. But in some sectors, copyrights and

Table 3.8 *Merchandise trade and intellectual property rights by level of economic development*

Dependent Variable:	Merchandise Imports (in real 2000 US dollars)		
	Developed Countries (Group 1)	Developing Countries (Group 2)	Least Developed Countries (Group 3)
Constant	14.2*** (3.12)	10.6*** (1.82)	13.2*** (2.88)
Patent Rights Index	9.86*** (0.93)	1.34*** (0.16)	0.54* (0.31)
Real GDP per capita	0.09 (0.21)	0.56*** (0.088)	2.18*** (0.22)
Freedom to Trade Internationally	-2.45*** (0.66)	1.56*** (0.41)	-0.64* (0.37)
Doing Business Rank	0.17** (0.069)	-0.027 (0.07)	-0.80** (0.39)
IPR Survey	5.26*** (1.38)	3.31*** (0.49)	-0.82 (0.55)
Legal Effectiveness	0.91 (0.70)	0.37 (0.27)	-0.10 (0.22)
Physical Property Rights	-5.03** (2.40)	-0.39 (0.60)	1.15 (0.72)
Governance	-1.06*** (0.36)	-1.20*** (0.19)	0.37** (0.16)
Number of Observations	69	164	31
Pseudo-R ²	66.9%	36.5%	74.0%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. Standard errors are in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels respectively. Group 1 consists of developed countries, Group 2 of developing countries, and Group 3 of least developed countries; see Appendix 3.1 for the detailed listing of countries.

trademark rights may be more important if they are the only legal means of protecting an intellectual property asset; for example, software may only be copyrightable, not patentable.

The negative coefficients of copyrights and trademark rights in column 4, though statistically insignificant, likely arise because all three indexes are positively correlated but the copyright and trademark variables are poorer proxies for IPRs that matter to FDI.¹⁹ On the whole, the estimates suggest that a 1% strengthening of patent rights is associated with a more than 2% increase in the stock of inward FDI. The model explains about 60%

Table 3.9 Service trade and intellectual property rights

Dependent Variable:	Service Imports (in real 2000 US dollars)			
	(1)	(2)	(3)	(4)
Constant	-2.18** (0.99)	-2.14*** (1.14)	-6.22*** (1.61)	-7.16*** (1.83)
Patent Rights Index	1.22*** (0.17)			1.93*** (0.29)
Copyrights Index		-0.16 (0.12)		-0.45** (0.23)
Trademark Rights Index			0.07 (0.15)	-0.45** (0.19)
Real GDP per capita	0.79*** (0.047)	0.85*** (0.056)	0.87*** (0.077)	0.72*** (0.10)
Freedom to Trade Internationally	-0.17 (0.32)	0.41 (0.34)	1.66*** (0.52)	1.73*** (0.53)
Doing Business Rank	-0.049 (0.056)	-0.007 (0.053)	0.079 (0.084)	0.13 (0.08)
IPR Survey	2.75*** (0.37)	3.37*** (0.44)	3.86*** (0.49)	4.79*** (0.65)
Legal Effectiveness	0.79*** (0.20)	0.26 (0.23)	0.77** (0.30)	0.60* (0.34)
Physical Property Rights	-0.89** (0.41)	-1.14** (0.49)	-1.29** (0.59)	-2.49*** (0.77)
Governance	-1.21*** (0.14)	-1.05*** (0.15)	-1.38*** (0.18)	-1.34*** (0.20)
Number of Observations	247	225	178	166
Pseudo-R ²	64.1%	58.7%	59.4%	63.6%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. Standard errors are in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels respectively.

of the variation in the data.²⁰ Of course, some of this FDI could be for the establishment of plants whose main purpose is sales and distribution or manufacturing, rather than research. The fact that multinational firm investments are sensitive to patent rights, holding other factors constant, suggests that there are valuable intangible assets at stake, whether they are for production, research, or sales.

As for the control variables, real GDP per capita is a statistically significant explanatory variable as is the index of free trade. The variable, IPR survey, which reflects business executives' perceptions (or experiences) of IPR strength is also a significant determinant. Legal effectiveness in

Table 3.10 *Service trade and intellectual property rights by level of economic development*

Dependent Variable:	Service Imports (in real 2000 US dollars)		
	Developed Countries (Group 1)	Developing Countries (Group 2)	Least Developed Countries (Group 3)
Constant	-8.02*** (3.15)	-1.32 (2.07)	-0.31 (3.32)
Patent Rights Index	9.99*** (0.88)	0.99*** (0.19)	0.97** (0.40)
Real GDP per capita	0.39* (0.21)	0.64*** (0.10)	2.12*** (0.22)
Freedom to Trade Internationally	-2.21** (0.87)	0.54 (0.58)	-1.68*** (0.49)
Doing Business Rank	0.10 (0.07)	-0.15 (0.11)	-0.72 (0.46)
IPR Survey	0.18 (1.25)	3.98*** (0.53)	1.15 (0.91)
Legal Effectiveness	0.98 (0.98)	0.87*** (0.29)	0.28 (0.31)
Physical Property Rights	1.72 (2.04)	-2.34*** (0.64)	-0.81 (1.12)
Governance	-1.05*** (0.41)	-1.31*** (0.22)	-0.06 (0.28)
Number of Observations	67	153	27
Pseudo-R ²	68.1%	33.3%	69.9%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. Standard errors are in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels respectively. Group 1 consists of developed countries, Group 2 of developing countries, and Group 3 of least developed countries; see Appendix 3.1 for the detailed listing of countries.

general is not consistently an important factor in inward FDI. This could be due to the possibility that investors are more concerned with specialized legal protection, as it relates to intellectual property, rather than general property rights and legal institutions in general. The doing business variable has the right sign (negative, since lower scores reflect higher rank) but is not statistically significant, meaning that the cost of starting and operating a business is conducive to FDI but is not a primary influence. Governance has a statistically significant negative influence on inward FDI but this stems from two factors. First, governance and intellectual property rights

Table 3.11 US FDI operations in developing countries and intellectual property rights by industry

Dependent Variable:	Total Assets of Foreign Affiliates of US Multinational Firms (in real 2000 US dollars)					
	(1) Chemicals	(2) Machinery	(3) Electrical appl. & components	(4) Service	(5) Computers & electronics	(6) Information
Constant	-6.54* (3.92)	-0.02 (5.19)	-6.81 (4.74)	-2.37 (3.45)	-9.32 (7.31)	-2.24 (5.75)
Patent Rights Index	1.14*** (0.37)	1.40*** (0.49)	-0.43 (0.51)	1.74*** (0.32)	0.96 (1.49)	5.93*** (0.64)
Real GDP per capita	0.43* (0.06)	0.33 (0.27)	0.39 (0.30)	0.45*** (0.17)	-0.29 (0.54)	0.48 (0.30)
Freedom to Trade Internationally	-3.98*** (1.29)	-1.97 (1.49)	0.43 (1.45)	-0.26 (1.27)	1.78 (3.31)	-4.05*** (1.23)
Doing Business Rank	-0.33** (0.15)	-0.60* (0.31)	0.096 (0.26)	-0.46*** (0.12)	-0.25 (0.39)	-0.04 (0.24)
IPR Survey	3.03*** (1.23)	3.61*** (1.62)	4.57*** (2.11)	2.66*** (1.03)	12.1*** (3.88)	2.85* (1.68)
Legal Effectiveness	0.94 (0.62)	3.29*** (0.78)	1.85* (1.02)	0.18 (0.81)	3.55*** (0.78)	4.83*** (0.72)
Physical Property Rights	-1.09 (1.45)	-2.15 (1.55)	-0.95 (1.76)	0.31 (1.12)	-5.31 (3.63)	-5.26*** (1.39)
Governance	-0.83*** (0.35)	-2.64*** (0.41)	-0.93* (0.54)	-1.32*** (0.41)	-1.78** (0.86)	-0.71 (0.68)
Number of Observations	81	59	55	76	41	49
Pseudo-R ²	25.8%	22.1%	21.2%	30.8%	32.3%	48.7%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. All columns in the table are mutually exclusive. Standard errors are in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels respectively.

Table 3.12 Merchandise trade in developing/least developed countries and intellectual property rights by commodity group

	Dependent Variable: Merchandise Imports (in real 2000 US dollars)					
	(1) Pharmaceuticals	2) Office Telecom	(3) Chemicals	(4) Optics	(5) Electronics	(6) Aerospace
Constant	9.55*** (1.39)	8.13*** (1.14)	3.71** (1.65)	7.20*** (1.34)	7.54*** (1.47)	2.94 (2.98)
Patent Rights Index	1.18*** (0.21)	1.45*** (0.21)	1.26*** (0.15)	1.29*** (0.17)	1.54*** (0.19)	2.03*** (0.36)
Real GDP per capita	0.85*** (0.07)	1.03*** (0.072)	1.07*** (0.08)	0.96*** (0.06)	0.95*** (0.062)	0.80*** (0.15)
Freedom to Trade Internationally	-0.55 (0.52)	0.58 (0.37)	0.19 (0.51)	-0.13 (0.44)	0.78* (0.44)	0.61 (0.97)
Doing Business Rank	0.09* (0.05)	-0.27*** (0.081)	0.11* (0.07)	-0.08 (0.06)	-0.32*** (0.07)	-0.44*** (0.14)
IPR Survey	0.56 (0.46)	2.35*** (0.55)	3.97*** (0.45)	2.65*** (0.26)	3.48*** (0.41)	3.54*** (0.79)
Legal Effectiveness	-0.09 (0.31)	0.05 (0.25)	0.56* (0.31)	0.55** (0.24)	0.51** (0.25)	1.29** (0.59)
Physical Property Rights	1.51*** (0.53)	0.053 (0.59)	-0.19 (0.55)	-0.21 (0.25)	-0.54 (0.48)	0.25 (1.14)
Governance	-1.03*** (0.17)	-0.91*** (0.19)	-1.96*** (0.20)	-1.02*** (0.16)	-1.59** (0.16)	-2.07*** (0.39)
Number of Observations	124	172	162	162	162	161
Pseudo-R ²	54.0%	54.6%	47.2%	53.6%	54.9%	32.0%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. All columns in the table are mutually exclusive. Standard errors are in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels respectively.

Table 3.13 Trade in services in developing and least developed countries and intellectual property rights by sector

Dependent Variable:	Service Imports (in real 2000 US dollars)		
	Communications (1)	Computers (2)	Royalties (3)
Constant	-9.01*** (1.67)	-14.7*** (2.56)	-6.10** (2.63)
Patent Rights Index	1.55*** (0.27)	3.32*** (0.34)	1.73*** (0.31)
Real GDP per capita	0.67*** (0.08)	0.97*** (0.11)	0.97*** (0.12)
Freedom to Trade Internationally	1.75*** (0.53)	0.43 (0.86)	0.02 (1.45)
Doing Business Rank	0.06 (0.14)	0.40** (0.18)	-0.38** (0.16)
IPR Survey	2.62*** (0.59)	2.38*** (0.79)	5.26*** (0.89)
Legal Effectiveness	0.30 (0.31)	1.19*** (0.46)	1.01** (0.44)
Physical Property Rights	-0.89 (0.62)	-0.53 (0.78)	-3.95*** (1.07)
Governance	-0.88** (0.21)	-1.41*** (0.29)	-1.67*** (0.27)
Number of Observations	142	91	136
Pseudo-R ²	43.3%	41.9%	42.3%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. Standard errors are in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels respectively.

are positively correlated such that when the two variables are entered together, the positive effect of IPR variables dominates. Second, the net negative influence could suggest some substitution effects (although the result should be interpreted with some caution due to the aforementioned correlation); that stronger governance favours domestic capital formation rather than foreign investment, such that domestic capital crowds out some inward FDI.

In Table 3.6, the results of dividing the sample into different country groups are shown. This provides an indication whether the effects of IPR on inward FDI vary by level of economic development. To conserve space, only the results with patent rights are shown since patent protection is the more important intellectual property factor concerning inward FDI. As the table shows, the positive relationship of IPRs to inward

Table 3.14 Research and development (R&D) and intellectual property rights in developing economies

Dependent Variable:	R&D as a percentage of GDP			
	(1)	(2)	(3)	(4)
Constant	-8.01*** (0.68)	-9.48*** (0.81)	-8.1*** (0.66)	-11.1*** (1.35)
Patent Rights Index	1.04*** (0.16)	0.77*** (0.15)	0.76*** (0.13)	0.87*** (0.14)
Inward FDI Stock	-0.03 (0.02)			-0.12*** (0.028)
Merchandise Imports		0.11*** (0.035)		0.22*** (0.08)
Service Imports			0.09** (0.04)	0.04 (0.08)
Quality of Research Institutions	1.90*** (0.43)	2.02*** (0.38)	2.11*** (0.37)	1.49*** (0.36)
University-Industry Research Collaboration	2.37*** (0.39)	1.49*** (0.41)	1.65*** (0.40)	1.52*** (0.36)
Other Institutional Controls (see Notes)	Yes	Yes	Yes	Yes
Number of Observations	131	131	127	127
Pseudo-R ²	56.0%	57.3%	56.9%	58.6%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. Standard errors are in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels respectively. The sample here consists of developing and least developed countries.

Other Institutional Controls include 'Doing Business Rank', 'IPR Survey', 'Physical Property Rights', 'Legal Effectiveness', and 'Governance'. The coefficient estimates and standard errors for these variables are not reported in order to conserve space. The results are available from the authors upon request.

FDI is present for all three country income groups. In each case, the inward FDI stock is positively and significantly related to stronger patent protection. However, the quantitative relationship does vary by level of development. The impact is largest among developed countries. A 1% strengthening of patent laws is associated with an 11.2% increase in the stock of inward FDI in Group 1, but with a less than 2% increase among Groups 2 and 3 countries.²¹ This may be due to the existence of complementary factors in Group 1 that make FDI more profitable in developed country markets.

The IPR survey variable also shows that business perceptions of a stronger intellectual property regime have a stronger impact on inward

Table 3.15 Residential patenting and intellectual property rights in developing economies

Dependent Variable:	Patent Applications by Residents			
	(1)	(2)	(3)	(4)
Constant	-12.0*** (1.26)	-28.0*** (1.83)	-12.5*** (1.76)	-28.9** (4.23)
Patent Rights Index	1.95*** (0.35)	0.84** (0.34)	1.16*** (0.31)	1.09*** (0.31)
Inward FDI Stock	0.49*** (0.052)			-0.24** (0.12)
Merchandise Imports		1.08*** (0.084)		1.11*** (0.28)
Service Imports			1.13*** (0.09)	0.39 (0.26)
Quality of Research Institutions	7.56*** (0.43)	5.98*** (1.04)	5.22*** (0.91)	5.32*** (1.22)
University-Industry Research Collaboration	1.20 (1.07)	-0.29 (1.25)	0.85 (1.24)	-0.51 (1.59)
Other Institutional Controls (see Notes)	Yes	Yes	Yes	Yes
Number of Observations	112	112	103	103
Pseudo-R ²	61.1%	68.6%	66.5%	70.2%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. Standard errors are in parentheses. ***, **, * indicate statistical significance at the 1%, 5%, and 10% levels respectively. The sample here consists of developing and least developed countries.

Other Institutional Controls include 'Doing Business Rank', 'IPR Survey', 'Physical Property Rights', 'Legal Effectiveness', and 'Governance'. The coefficient estimates and standard errors for these variables are not reported in order to conserve space. The results are available from the authors upon request.

FDI in Group 1 than in Group 2. The estimated coefficient of IPR survey is 6.88 for Group 1 versus 1.63 for Group 2. However, the coefficient of IPR survey for Group 3 is negative. There are three possible reasons. For Group 3 countries, again the survey and the index are positively correlated, but what matters in LDCs is not just the perception of strength but actual legislative changes and statutory rights which give a clear signal to agents as to what their intellectual property rights are and what means of enforcement exist. Also, the IPR survey reflects not just patent strength but intellectual property rights in general, including copyrights and trademark rights. The net negative effect may then reflect the possibility that strong, comprehensive IP enforcement exerts some market power effects. Firms

Table 3.16 *Non-residential patenting and intellectual property rights in developing economies*

Dependent Variable:	Patent Applications by Non-Residents			
	(1)	(2)	(3)	(4)
Constant	-5.09*** (0.92)	-20.5*** (1.09)	-4.85*** (0.65)	-11.7*** (2.33)
Patent Rights Index	1.01*** (0.19)	1.11*** (0.21)	1.57*** (0.17)	1.05*** (0.17)
Inward FDI Stock	0.79*** (0.034)			0.19*** (0.07)
Merchandise Imports		1.03*** (0.051)		0.47*** (0.14)
Service Imports			1.14*** (0.045)	0.50*** (0.15)
Quality of Research Institutions	2.13*** (0.56)	0.08 (0.54)	-0.59 (0.56)	0.79 (0.56)
University-Industry Research Collaboration	1.84*** (0.66)	0.82* (0.05)	1.02** (0.46)	0.07 (0.57)
Other Institutional Controls (see Notes)	Yes	Yes	Yes	Yes
Number of Observations	125	125	115	115
Pseudo-R ²	72.5%	76.2%	74.6%	77.1%

Notes: All variables are in natural logarithmic units, except for Governance. See Appendix 3.2 for variable definitions and data sources. Standard errors are in parentheses. *** ** * indicate statistical significance at the 1%, 5%, and 10% levels respectively. The sample here consists of developing and least developed countries.

Other Institutional Controls include 'Doing Business Rank', 'IPR Survey', 'Physical Property Rights', 'Legal Effectiveness', and 'Governance'. The coefficient estimates and standard errors for these variables are not reported in order to conserve space. The results are available from the authors upon request.

that enjoy increased market power have an incentive to reduce the stock of inward FDI and consequently the supply of output of plants.

As for the control variables, GDP per capita or income per person has little predictive value for the inward FDI of developed countries. The high levels of GDP per capita in richer countries may reflect relatively high labour costs which in some cases create offsetting incentives to increase the stock of inward FDI that higher income per person might otherwise generate. The index of free trade is also a weak influence on FDI in Group 1. Generally, Group 1 countries all have open markets so that this variable exhibits little variability among developed economies. The other institutional variables are not consistently important factors and they are,

as pointed out earlier, correlated with IPRs. The important demonstration here is that the patent rights variable remains statistically significant even after controlling for these other institutional factors – factors which some might argue are what the intellectual property rights indexes really pick up. Instead the results indicate that IPRs have influences beyond any institutional aspects that are correlated with strong IPRs. These influences specifically affect incentives to transfer knowledge-related assets which are easy to imitate and difficult to appropriate.

Judging by the goodness of fit measure (pseudo- R^2), the model overall does not fit Group 2 as well as it does for Groups 1 and 3. Only a third of the variation in the inward FDI of Group 2 is captured by the model. There are a lot more countries in Group 2 and more heterogeneity among them. The group consists of new emerging markets like Bulgaria, the Czech Republic, Slovakia, Hungary, and Poland, newly industrialized countries like South Korea, Chinese Taipei, Singapore, and Hong Kong, as well as lower income economies like Egypt, Sri Lanka, and Pakistan.

Table 3.7 repeats the analysis displayed in Table 3.5 with merchandise imports as the variable to be explained. The results show that patent rights are a positive, statistically significant determinant of merchandise imports. Copyrights are an insignificant influence and trademark rights a mild influence on merchandise imports. But overall the effects of patent rights dominate. Enforcement of IPRs in practice is also a significant influence on inflows of merchandise imports. Other factors like the freedom to trade and GDP per capita also positively and significantly influence merchandise imports. Again, patent rights are an important determinant even after controlling for related institutional variables, like legal effectiveness, governance, property rights in general, and the business environment. Around 60% of the variation in merchandise imports is accounted for by the model.

Table 3.8 shows the results of splitting the sample by country groups. Patent rights are a statistically significant determinant of merchandise imports for both developed and developing countries; however, the quantitative impact of a percentage change in patent reform is larger for developed countries than it is for developing. For LDCs, the qualitative and quantitative impacts of patent rights on merchandise imports are weaker. The patent rights variable is statistically significant at only the 10% level and a 1% increase in the strength of patent protection is associated with less than a 1% increase in merchandise imports. Moreover, the IPR survey variable is largely irrelevant to the merchandise imports of LDCs. One possibility is that stronger IPR enforcement prevents continued importation of cheap imitations which in some cases might offset increases of other types of merchandise imports.

As for the control variables, income per person positively affects the demand for merchandise imports in Groups 2 and 3 but negligibly in Group 1. The ability to engage in free trade positively influences the merchandise imports of developing countries but may induce substitution effects in developed countries and in LDCs; that is, encourage a switch from merchandise imports to inward FDI. Overall the model explains more than two-thirds of the variation in merchandise imports in Groups 1 and 3 but only one-third of it in Group 2.

Table 3.9 shows how service imports relate to intellectual property protection and other variables. Columns 1–3 examine the different indexes of IPRs one at a time while column 4 examines all of them jointly. Again, patent rights are a dominant influence in that service imports are more sensitive to levels of patent protection, holding other variables constant. Most of the control variables (index of free trade, GDP per capita, IPR enforcement, and legal effectiveness) exert a positive influence on service imports. About 60% of the variation in service imports is explained by the model.

Table 3.10 shows the results of splitting the sample by country groups. Again, the positive and statistically significant influence of patent rights on service imports does not vary by level of economic development, but the quantitative impact on such imports is largest in richer countries. The enforcement of IPRs in practice is important in developing countries, but not significant in the developed and least developed markets. In those regions, the market expansion effects of stronger IPR laws and enforcement may be balanced by the increased market power effects. The model explains almost 70% of the variation in the data for developed and least developed countries but only a third for developing countries. Again, the middle sample is more heterogeneous.

5.2 Technology Content of Transfers in Developing Countries

The results thus far have established a positive association between patent rights and technology transfer, controlling for other variables, and that this association is not qualitatively different across different income groups of countries. This supports the view that the recent increase in developing country trade and FDI can be attributed to the strengthening of IPRs in developing countries, among other factors. Next, the technological content and technological implications of the increased trade and FDI are assessed. For example, are there increased inflows of technologically-intensive goods, services, and capital in developing countries? Do these inflows stimulate R&D and patenting in developing countries by both residents and non-residents?

Table 3.11 examines US FDI expansion in developing countries by

sector. Stronger patent protection in developing countries is associated with a greater expansion of US operations in the chemical, machinery, service and information industries in developing countries. This underscores the important role of IPRs in the service industry. Stronger patent protection in developing countries, however, is not associated with a greater expansion of US operations in the electronic and computer industries of developing countries. It may be that US FDI in the electronics and computer industries of developing countries is largely for sales and distribution purposes, so that there is less sensitivity to the strength of local patent rights. Nonetheless, IPR enforcement in general does matter to all of the sectors considered (see the “IPR survey” variable), particularly to protect against piracy and unauthorized sales and distribution.

As for the control variables, the doing business variable and legal effectiveness generally promote sectoral FDI. Greater freedom to trade may exert some substitution effects in the chemical and information sectors, encouraging firms to supply developing country markets via trade rather than subsidiary production. At the sectoral level, the model does not explain a large proportion of the variation in the data. In some cases, only a fifth of the variation is explained.

Table 3.12 examines the merchandise imports of developing countries by sector. Patent protection is strongly significant and positively associated with imports of pharmaceutical goods, chemicals, office and telecom equipment, electronics, aerospace, and optics and precision equipment. Thus, goods that are technologically-intensive are particularly affected by stronger patent protection in developing countries. It is interesting that the “IPR Survey” variable matters to all the sectors but pharmaceuticals. This suggests that for imports of drugs, strong statutory protection matters more than perception of strength or enforcement. It may be argued that specific legislations give patent owners clear and stronger bargaining power vis-à-vis imitators or unauthorized importers or producers. About 50% of the sectoral variation of merchandise imports is explained by the model, except for aerospace goods and services, for which the model explains about a third of the variation.

Table 3.13 examines the service imports of developing countries by select sectors. Patent protection and IPR enforcement are both statistically significant at the 1% level and are positively associated with the imports of communication service, computer service, and royalties. The increased royalty payments represent the increased use of foreign intangible assets by the developing world. Thus the results suggest that stronger intellectual property protection helps attract imports of services that are knowledge-related or technology-intensive. This holds even after controlling for GDP per capita and other institutional factors.

The next set of analyses takes a different tack. Assuming that inward FDI and imports bring new technological knowledge or products into developing countries, the host countries could experience increased innovation potential. Local researchers may benefit from knowledge spillovers or increased access to laboratory equipment, components, and research materials like chemicals, computer programs, and biological samples. Stronger patent protection in developing countries could enhance local R&D directly by increasing the ability to appropriate the returns to R&D, and indirectly by stimulating imports of goods and services and inward FDI. In addition, increased innovation outputs would be associated with an increase in resident patenting. Furthermore, to the extent that imports and inward FDI embody new technological processes or products, foreign firms are likely to apply for local patents. Thus technological inflows would be associated with a rise in non-resident patenting. Stronger patent protection in developing countries could therefore affect patenting directly (by increasing the strength of patent rights) as well as indirectly by stimulating the inflow of patent-sensitive imports or FDI.

The results in Table 3.14 show that R&D in developing countries is enhanced by stronger patent rights. Moreover, merchandise imports positively explain local R&D. Service imports seem to affect local R&D but are an insignificant influence once merchandise imports are controlled for. The stock of inward FDI is not a significant, positive influence on local R&D. This would be the case if developing country FDI is largely for purposes of sales, distribution, or manufacturing in order to take advantage of lower wages and not for research and development per se. The reason merchandise trade may matter more than service trade to local R&D is that merchandise imports provide tangible goods, like equipment, machinery, and materials, for conducting R&D. Service imports provide important know-how and tacit knowledge, but merchandise goods appear to be more influential for developing country research.

In Table 3.14, the regression model controls for not only the institutional variables previously considered, such as legal effectiveness, but also those institutional factors that should be especially relevant to innovation, for example the extent of university-industry research collaboration and the quality of local research institutions, both of which are based on surveys of business executives. Both variables are indeed statistically significant determinants of local R&D.

Table 3.15 presents results on resident patenting in developing countries. Patent protection is a statistically significant determinant of patent applications filed by residents. Moreover, each of the modes of technology transfer – inward FDI, merchandise imports, and service imports – is also a significant determinant of resident patenting when considered individu-

ally. However, when all three modes are considered jointly, merchandise imports dominate. The reasoning is similar to that above. Imports of equipment and materials may help to stimulate R&D, and to the extent that increased R&D results in increased innovative outputs for which local residents file for patent protection, increased resident patenting will be positively associated with merchandise imports. Note that the quality of research institutions is important for resident patenting but university-industry collaboration is not a statistically significant determinant. The latter collaboration may be more important in the R&D stage where firms and academia can collaborate on basic, foundational research. At the patenting stage, firms may want to keep commercial successes proprietary and rely less on collaboration with academia. Furthermore, in most developing nations, US Bayh-Dole type laws (allowing or encouraging publicly funded university research outputs to be patented) do not exist.

Table 3.16 shows that patent protection is positively and significantly associated with non-resident patenting in developing countries, holding other factors constant. Non-resident patent filings are also positively and significantly associated with all three modes of technology transfer – inward FDI, merchandise imports, and service imports – even when all three modes are entered jointly. This suggests an important degree of patent sensitivity of foreign imports and foreign capital in developing markets. If non-residents did not bring a significant amount of new technologies to these markets, they would not pursue patent applications in the host markets. Transferring old vintages obviates the need to file costly patent applications. Thus, the positive relationship between non-resident patenting and foreign imports and capital should reveal something about the technological content of the imports of goods, services and capital of these countries.

Turning to the control variables, it can be seen that the quality of local research institutions is generally not a significant determinant of non-resident patenting since the R&D is conducted elsewhere. The university-industry research collaboration is significant only when one of the technology transfer modes is modelled. When all three modes are considered, the variable loses its robustness. The research collaboration variable may be picking up the demand for technology transfer or exchanges such that when all three modes of technology transfer are controlled for, the variable is no longer significant.

5.3 Summing Up

The strength of patent protection is positively associated with inward FDI, merchandise imports, and service imports. This relationship holds

for all groups of countries – developed, developing, or least developed – though quantitatively the association is strongest in developed countries. Copyrights and trademark rights matter less than patent rights. In addition, stronger levels of patent protection in developing countries are positively and significantly associated with the inflows of high-tech products, like pharmaceutical goods, chemicals, computer services, information, and office and telecom equipment. Those inflows of goods, services, and capital, in turn are positively associated with local R&D and patenting, as well as non-resident patenting. The latter association suggests that foreign imports and FDI contain patentable technological assets for which non-residents have a vested interest in seeking protection. Thus stronger patent rights in developing countries have the potential not only to attract technology transfer but also encourage foreigners to transfer new technologies.²² The evidence here is thus supportive of certain classes of theoretical analyses of North-South models that find mutual gains from IPR reform. For example, Naghavi (2007) develops a model in which stronger Southern IPRs can attract FDI into developing countries and stimulate local innovation.

6. CONCLUSION

Since the TRIPS Agreement came into force in 1995, developing countries and LDCs have undertaken significant reforms in their intellectual property systems. The IPR indexes show that between 1995 and 2005, developing and least developed countries, as a group, experienced a greater percentage increase in IPR strength than did the developed world. During the same period, compared to developed countries, the developing and least developed countries experienced large growth in inflows of inward FDI, merchandise and service imports, and patent applications by foreigners, as well as increases in their R&D to GDP ratios and patenting by local residents.

The empirical analysis examined the extent to which these developments can be attributed to intellectual property reforms in the developing world, holding other factors constant. Specifically the study evaluated two hypotheses regarding the experience of the developing world with strengthened IPRs:

H1: IPRs stimulate technology transfer, particularly the transfer of technology-intensive goods, services and capital.

H2: IPRs can directly stimulate local innovation as well as indirectly by stimulating the transfer of technologies that foster local innovation.

The main empirical results support a positive assessment of both hypotheses. These results are as follows:

- The index for patent rights tends to be positively associated with inward FDI, merchandise imports and service imports, holding other factors constant. This relationship holds for all groups of countries – developed, developing and least developed – though quantitatively the association is strongest in developed countries.
- The indexes for copyrights and trademark rights are less strongly associated with technology transfer than is the patent rights index.
- Focusing on technology transfer to developing countries, the study finds that stronger levels of patent protection are positively and significantly associated with the inflows of high-tech products, like pharmaceutical goods, chemicals, aerospace, computer services, information, and office and telecom equipment.
- Developing country patent applications (by both residents and non-residents) and expenditure on R&D (as a percentage of GDP) tend to have a positive and significant relationship to the strength of patent rights.
- In certain specifications of the model with respect to developing country resident patent applications, the control variables for merchandise imports, services imports and inward FDI are also significant and positive. Similarly, merchandise imports also tend to be significantly and positively related to R&D expenditure. (When FDI and service imports are interacted with merchandise imports in these specifications of the model, merchandise imports tend to dominate, however.) The intuition is that the inflows of goods, services and capital are a source of knowledge spillovers and inputs to conduct innovation (such as laboratory equipment).
- Imports of goods and services and FDI inflows are also significantly and positively associated with non-resident patenting in developing countries. This association suggests that foreign imports and FDI contain patentable technological assets for which non-residents have a vested interest in seeking protection. Thus, stronger patent rights in developing countries appear to have the potential not only to stimulate international technology transfer but also to provide incentives for foreigners to transfer *new* technologies.

At the same time, it should be noted that IPRs alone do not determine technological success or even increased access to technology. Indeed, regression analysis is conditional on other factors being held constant. It is important to note that IPRs do not operate in a vacuum. There are

complementary factors that help facilitate technology transfer, such as the quality of infrastructure, government policies and regulations, and market structure, among others. There are also complementary factors that specifically affect innovation and technology diffusion, such as the quality of knowledge institutions (e.g. academia, public research institutes or industrial research centers like science parks), the financial system (e.g. access to equity, credit and venture capital), availability of trained human capital, and networks for research collaboration or interaction (e.g. university-industry research collaboration or international collaboration).

In terms of a future research agenda, it would be useful to explore the requisite complements to IPR reform, as part of an integrated policy for development of domestic technological capacity. What are the underlying technical, institutional, and policy determinants of innovative and absorptive capacities? What role does the multilateral trading system play in shaping these determinants? What feedback mechanisms exist with respect to IPR strengthening? In particular, what influence do IPRs have, if any, on opportunities and incentives for agents to acquire greater capabilities or for firms to aggressively seek and absorb new technologies including via international trade and investment.

NOTES

1. Of course, this point must be qualified. Firms may also have other means of protecting their assets, such as trade secrecy, copyright protection, reputation, and lead time. Furthermore, technologies that are hard to replicate provide some de facto protection.
2. See Park and Lippoldt (2003, 2005) for a review of the academic and policy debates.
3. E.g. Liu and Lin (2005) using a dataset covering 1989–2000 find no evidence of a structural change in the relationship between Taiwanese exports and foreign patent rights after TRIPS is implemented.
4. At the 2001 Doha Ministerial Conference, WTO members issued the *Declaration on The TRIPS Agreement and Public Health*, to make clear their intention for the TRIPS Agreement to contribute positively to public health; available here: http://www.wto.org/english/thewto_e/minist_e/min01_e/mindecl_trips_e.htm.
5. Finger and Schuler (1999) provide an assessment of costs related to the implementation of WTO Uruguay Round commitments with respect to IPR reform, among other issues.
6. The criticism is certainly not limited to developing country observers. Scholars such as L. Lessig and M. Boldrin and D. K. Levine have questioned the strengthening of IPRs more generally. E.g. Lessig has challenged the privatization of the so-called “intellectual commons” (Lessig, 2002) and the expanded range of patentable innovation in the US that now includes such areas as Internet business methods (Lessig, 1999). Boldrin and Levine (2007) recognize the need for innovators to be rewarded and to have the right of sale with regard to their ideas. However, they challenge the right to regulate the use of innovations after their sale, proposing instead that innovators should make better efforts to capitalize on their first-mover advantages when they first sell their ideas.
7. Expressing similar concerns, in 2004 Brazil and Argentina proposed for WIPO to launch a new development agenda. The ensuing debate led to an evolution of the concept and

- eventual endorsement of a development agenda by the WIPO General Assembly on 28 September 2007: http://www.wipo.int/pressroom/en/articles/2007/article_0071.html.
8. E.g., Cambodia, China and Singapore have integrated IPRs in their national economic strategies and affirmed the importance of IPRs on their national intellectual property office web sites: http://www.camcl.org/sub/training_materials/intellectual_property-rights.html, http://www.sipo.gov.cn/sipo_English/laws/developing/200906/t20090616_465239.html, <http://www.ipos.gov.sg/topNav/abo/Vision+Mission+and+Values.htm>.
 9. See Blonigen (2001) for an analysis of the substitution effects or complementarities between a multinational firm's exports and production by its foreign affiliates, using product-level data.
 10. Mayer and Pfister (2001) also obtain an inverse-U relationship for developing countries if the level of corruption and political rights in the host country are not controlled for. They argue that patent indexes may be picking up the effects of institutional factors and the regulatory climate.
 11. Another dimension of this issue is at which point this may matter for development. To the extent that older vintages of technology increase productivity and growth, they can still promote increased incomes.
 12. The commercial age of a technology is defined as the time from commercial introduction to the inception of a transfer agreement.
 13. The analysis here considers the *association* of change in indicators for the strength of intellectual property rights with changes in key economic variables, controlling for other factors. It does not explicitly measure *causality*. Nevertheless, IPR reforms are generally signaled in advance and, in effect, are likely to have preceded the changes recorded in the dependent economic variables considered in the regression analysis. Later we discuss a further test using lagged dependent variables.
 14. See Taylor (1994) and Grossman and Lai (2004) for a theoretical analysis of the linkages among IPRs, innovation, and openness.
 15. A discussion of technology and globalization indicators can be found in OECD (2005).
 16. The country grouping is based on a UN classification: UNCTAD *Handbook of Statistics* (2005), pp. xi–xv.
 17. The industrial categories in the BEA data have changed since 1999 so that some sectoral matching with prior years is difficult.
 18. The results of the regression assessment simultaneously taking into account all three IPR indexes should be interpreted with caution due to potential multicollinearity issues, discussed below.
 19. Typically if two variables are positively correlated and are entered together in a model, their coefficient signs may be opposite. See Greene (2003), Chapter 4.9.
 20. $Pseudo R^2 = (corr(y, y^*))^2$, where y^* is the predicted values of the dependent variable y and $corr$ denotes correlation.
 21. FDI inflows normally respond to changes in the host country policy environment with a lag, e.g. due in part to the time it takes to plan and negotiate an investment. On the other hand, IPR reform in a given country generally rolls out over a multi-year period and hence there is an advance signalling effect to investors of the intentions of the government even before the laws on the books (and hence the IPR index scores) change. Thus, the model naturally includes a degree of built-in lag that may offset the effect of the stickiness of FDI. Nevertheless, in order to test whether there is a need for additional steps in the model to account for the lag in FDI response, the model was re-run with a lag of one five-year period. Overall, and for groups 1 and 2, the patent rights index was still strongly significant, although the significance was reduced for group 3 (LDCs).
 22. Recall that technologies are not patentable unless they are *novel* (i.e. new and not part of prior art as is the case for older technologies).

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APPENDIX 3.1

Table 3A.1 Sample of countries by group

Group 1: Developed Countries	Group 2: Developing Countries	Group 3: Least Developed
Australia	Algeria	Korea
Austria	Argentina	Lithuania
Belgium	Bolivia	Malaysia
Canada	Botswana	Mauritius
Denmark	Brazil	Mexico
Finland	Bulgaria	Morocco
France	Cameroon	Nicaragua
Germany	Chile	Nigeria
Greece	China	Pakistan
Iceland	Colombia	Panama
Ireland	Congo	Paraguay
Israel	Costa Rica	Peru
Italy	Cyprus ^a	Philippines
Japan	Czech Rep.	Poland
Luxembourg	Dominican Rep.	Romania
Malta	Ecuador	Russian Fed.
Netherlands	Egypt	Saudi Arabia
New Zealand	El Salvador	Singapore
Norway	Gabon	Slovak Rep.
Portugal	Ghana	Somalia
Spain	Grenada	South Africa
Sweden	Guatemala	Sri Lanka
Switzerland	Guyana	Swaziland
United Kingdom	Honduras	Syria
United States	Hong Kong	Chinese Taipei
	Hungary	Thailand
	India	Trinidad & Tobago
	Indonesia	Tunisia
	Iran	Turkey ^b
	Iraq	Ukraine
	Ivory Coast	Uruguay
	Jamaica	Venezuela
	Jordan	Vietnam
	Kenya	Zimbabwe

Notes:

- a The Republic of Cyprus is recognized by all members of the United Nations with the exception of Turkey. The information in this document relates to the area under the effective control of the Government of the Republic of Cyprus.
- b The information in this document with reference to "Cyprus" relates to the southern part of the Island. There is no single authority representing both Turkish and Greek Cypriot people on the Island. Turkey recognizes the Turkish Republic of Northern Cyprus (TRNC). Until a lasting and equitable solution is found within the context of United Nations, Turkey shall preserve its position concerning the "Cyprus issue".

APPENDIX 3.2 VARIABLES AND DATA SOURCES

A. Dependent Variables

1. Inward Foreign Direct Investment (FDI) Stock. Source: United Nations Conference on Trade and Development (UNCTAD) Foreign Direct Investment Database (FDI-online). <http://stats.unctad.org/fdi>
2. US Foreign Direct Investment Assets. Source: US Department of Commerce, Bureau of Economic Analysis, Financial and Operating Data Interactive Tables: <http://www.bea.gov/international/index.htm#omc>, "Total Assets of Majority-Owned Foreign Affiliates" of U.S. Multinationals, by Country and Industry for the years 1997, 2000, and 2004.
3. Merchandise Trade. Source: World Trade Organization (WTO), *Statistics Database, Time-series on Merchandise and Commercial Services Trade*, Subject Selection: Total Merchandise Trade. <http://stat.wto.org/StatisticalProgram/WSDBStatProgramSeries.aspx?Language=E>
4. Merchandise Trade by Sector. Sources:
 - Imports of Pharmaceuticals and Office and Telecom Equipment are from the WTO, *Statistics Database, Time-series on Merchandise and Commercial Services Trade*, Subject Selection: Merchandise Trade by Commodity: <http://stat.wto.org/StatisticalProgram/WSDBStatProgramHome.aspx?Language=E>.
 - Imports of Chemicals (Organic plus Inorganic), Electrical and Electronic Products, and Optics and Precision Equipment are from the United Nations *Comtrade Database*, Metadata and Reference: Commodity List: <http://comtrade.un.org/db/mr/rfCommodities-List.aspx>. Harmonized System Codes for these sectors are 28 (Inorganic chemicals), 29 (Organic Chemicals), 85 (Electricals and Electronics), 88 (Aircraft, Spacecraft), and 90 (Optics etc.).
5. International Trade in Services. Source: UNCTAD *Handbook of Statistics On-line*. <http://stats.unctad.org/handbook>. Report Folder V, Table 5.1.
6. International Trade in Services by Sector. Source: UNCTAD *Handbook of Statistics On-line*. <http://stats.unctad.org/handbook>. Report Folder V, Table 5.2.
7. Research and Development (R&D). Sources:

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 - Organisation for Economic Cooperation and Development (OECD), Science and Technology Database, *Main Science and Technology Indicators*, Vol. 2006, Rel. 02. www.sourceoecd.org.
8. Patent Applications (by Residents and Non-residents). Source: World Intellectual Property Organization (WIPO), *Industrial Property Statistics*, Table 1, various issues.

B. Explanatory Variables

1. Patent Rights Index. Source: Ginarte and Park (1997) and Park (2008).
2. Copyrights and Trademark Rights Indexes. Sources: Reynolds (2003) and Park (2005). [http://strategis.ic.gc.ca/epic/site/ippd-dppi.nsf/vwapj/09-EN%20Park.pdf/\\$file/09-EN%20Park.pdf](http://strategis.ic.gc.ca/epic/site/ippd-dppi.nsf/vwapj/09-EN%20Park.pdf/$file/09-EN%20Park.pdf)
3. IPR Survey. Source: World Economic Forum, *Global Competitiveness Reports 2000 (Table 7.09), 2004–5 (Table 6.04), 2005–6 (Table 6.04), and 2006–7 (Table 9.07)*, Oxford University Press. This variable measures Business Executive opinions on the stringency of intellectual property protection (1 = is weak or nonexistent, . . . , 7 = is equal to the world's most stringent).
4. Physical Property Rights. Source: World Economic Forum, *Global Competitiveness Reports 2000 (Table 3.11), 2004–5 (Table 6.03), 2005–6 (Table 6.03), and 2006–7 (Table 1.01)*, Oxford University Press. This variable measures Business Executive opinions on how well private property is protected, including financial assets (1 = are poorly defined and not protected by law, . . . , 7 = are clearly defined and well-protected by law).
5. Innovative Capacity. Source: World Economic Forum, *Global Competitiveness Reports 2000 (Table 7.01), 2004–5 (Table 9.04), 2005–6 (Table 8.03), and 2006–7 (Table 9.08)*, Oxford University Press. This variable measures Business Executive opinions on companies' capacity for innovation (1 = obtains technology by imitation of foreign technology or licensing, . . . , 7 = obtains technology by conducting formal research and pioneering their own new products and processes).

6. Quality of Research Institutions. Source: World Economic Forum, *Global Competitiveness Reports 2000 (Table 7.04)*, 2004–5 (*Table 3.05*), 2005–6 (*Table 3.05*), and 2006–7 (*Table 9.01*), Oxford University Press. This variable measures Business Executive opinions on the quality of scientific research institutions (1 = nonexistent, . . . , 7 = the best in their fields internationally).
7. University-Industry Research Collaboration. Source: World Economic Forum, *Global Competitiveness Reports 2000 (Table 7.06)*, 2004–5 (*Table 3.08*), 2005–6 (*Table 3.07*), and 2006–7 (*Table 9.03*), Oxford University Press. This variable measures Business Executive opinions on the extent to which businesses collaborate with local universities to engage in R&D (1 = minimal or nonexistent, . . . , 7 = intensive and ongoing).
8. Index of Legal Effectiveness. This variable is a composite score of judicial independence, impartial courts, security of property rights (tangible and intellectual), and integrity of the legal system. Source: Economic Freedom Network (EFN), 2006 Dataset: <http://www.free-theworld.com/release.html>.
9. Index of Freedom to Trade Internationally. This variable is a composite score of minimal taxes or interference with trade (including tariffs and non-tariff barriers to trade), minimal capital controls and foreign ownership restrictions. Source: Economic Freedom Network (EFN), 2006 Dataset: <http://www.freetheworld.com/release.html>.
10. Real GDP per capita (in real 2000 US dollars). Source: World Bank, *World Development Indicators (WDI) 2007*, CD-Rom.
11. Doing Business Rank. This variable measures the ease of doing business (for example, the number of procedures required in starting a business, dealing with licenses, employing workers, registering property, getting credit, protecting investors, enforcing contracts, paying taxes, trading across borders, and closing down a business). Countries are ranked in ascending order (1 = easiest place to do business, 2 = next easiest, etc.) Source: World Bank Group <http://www.doingbusiness.org/CustomQuery>.
12. Governance. Source: World Bank <http://info.worldbank.org/governance/wgi2007>.
There are six dimensions of governance: Voice and Accountability (VA), Political Stability (PS), Government Effectiveness (GE), Regulatory Quality (RQ), Rule of Law (RL), and Control of Corruption (CC). For purposes of this study, the six measures are averaged: Governance Index = $(VA + PS + GE + RQ + RL + CC)/6$

13. U.S. Price Index (GDP Deflator). Source: Economic Report of the President, <http://www.gpoaccess.gov/eop/download.html>.

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APPENDIX 3.3 INTELLECTUAL PROPERTY RIGHTS INDICES – SUMMARY OF CRITERIA AND MEASUREMENT

This appendix summarizes the components of each IPR index employed in the present study.

A. Patent Rights Index

Overall score for Patent Rights Index: sum of points under (1)–(5) in Table 3A2.

Table 3A.2 Patent rights index

(1) Membership in international treaties	Signatory	Not Signatory
Paris Convention and Revisions	1/5	0
Patent Cooperation Treaty	1/5	0
Protection of New Varieties (UPOV)	1/5	0
Budapest Treaty (Microorganism Deposits)	1/5	0
Trade-Related Intellectual Property Rights (TRIPS)	1/5	0
(2) Patentability coverage	Available	Not available
Pharmaceuticals	1/8	0
Chemicals	1/8	0
Food	1/8	0
Surgical products	1/8	0
Microorganisms	1/8	0
Utility models	1/8	0
Software	1/8	0
Plant & animal varieties	1/8	0
(3) Restrictions on patent rights	Does not exist	Exists
“Working” requirements	1/3	0
Compulsory licensing	1/3	0
Revocation of patents	1/3	–

Table 3A.2 (continued)

(4) Enforcement	Available	Not available
Preliminary injunctions	1/3	0
Contributory infringement	1/3	0
Burden-of-proof reversal	1/3	0
(5) Duration of protection	Full	Partial
	1	$0 < f < 1$ where f is the duration of protection as a <i>fraction</i> of 20 years from the date of application or 17 years from the date of grant (for grant-based patent systems).

B. Copyrights Index

Overall score for Copyright Index: average of (1)–(4) in Table 3A.3.

Table 3A.3 Copyrights index

(1) Coverage	
a. General (Literary and Artistic Works)	Duration as a percentage of 70 years
b. Performances	Duration as a percentage of 70 years
c. Sound Recordings	Duration as a percentage of 70 years
d. Films	Duration as a percentage of 70 years
e. Broadcasts	Duration as a percentage of 70 years
f. Droite de Suite (shares in resale)	Duration as a percentage of 70 years
g. Computer Programmes	Duration as a percentage of 70 years
Sub-Score (out of 1, average of a–g)	
(2) Usage (extent of private use)	Cumulative Score
i. Full use or no mention of private use	0
or ii. Private study or fair dealing	0.33
or iii. Use but with tax on devices or media	0.67
or iv. No private use allowed	1

Table 3A.3 (continued)

(3) Enforcement	Available	Otherwise
a. Criminal sanctions	1	0
b. Preliminary injunctions	1	0
c. Seizure and destruction	1	0
d. Anti-circumvention provision	1	0
Sub-score (out of 1, average of a–d)		
(4) International Treaties	Member	Otherwise
a. Berne Convention 1886	1	0
b. Universal Copyright Convention 1952	1	0
c. Rome Convention 1961	1	0
d. Geneva Convention 1971	1	0
e. Universal Copyright Convention 1971	1	0
f. Brussels Convention 1974	1	0
g. TRIPS Agreement 1995	1	0
Sub-score (out of 1, average of a – g)		

C. Trademark Rights Index

Overall score for Trademark Rights Index: average of (1)–(3) in Table 3A.4.

Table 3A.4 Trademark rights index

(1) Coverage	Available	Otherwise
a. Service marks	1	0
b. Certification marks	1	0
c. Collective marks	1	0
d. Colours	1	0
e. Shapes (3-dimensional, packaging, etc.)	1	0
f. Well-known marks	1	0
Sub-score (out of 1, average of a–f)	1	0
(2) Procedures	Available	Otherwise
a. Prohibition of marks in bona fide use	1	0
b. Licensing restrictions	1	0
c. Use or lose provisions in law	1	0
d. International exhibition protection	1	0

Table 3A.4 (continued)

e. Criminal penalties	1	0
f. Local lawyer requirements	1	0
g. Marks can become generic	1 (if law)	0
h. Transferability of mark without business	1 (if permitted)	0
i. Priority goes to first to use a mark	1 (if first to use)	0 (first-to-file)
Sub-score (out of 1, average of a – i)		
(3) International treaties	Member	Otherwise
a. Paris Convention 1883	1	0
b. Madrid Agreement 1891	1	0
c. Nice Agreement 1957	1	0
d. Lisbon Agreement 1958	1	0
e. Vienna Agreement 1973	1	0
f. Trademark Law Treaty 1994	1	0
g. TRIPS 1995	1	0
Sub-Score (out of 1, average of a–f)		